👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD MPP.7

Math is Beauty

Lesson 10-2 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use bilateral symmetry to describe structure in nature and in human creations.

Language Objective: I can describe a symmetric design using the words mirror image, line of symmetry, and balance.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

symmetric · **mirror image** · **bilateral symmetry** · **line of symmetry** · **balance**

1. Having two halves that match as mirror images is being _____.

2. A copy of a figure flipped across a line is its _____ _____.

3 Reflective symmetry in plants and animals is called _____ _____ .

4 The line you draw so each side reflects the other is the _____ _____ _____ .

5 An even arrangement where neither side feels heavier is _____ .

Write About the Math

How do you respond to your friend?
¿Cómo le respondes a tu amigo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **symmetric** — *simétrico*
- ☐ **mirror image** — *imagen reflejada*
- ☐ **bilateral symmetry** — *simetría bilateral*
- ☐ **line of symmetry** — *eje de simetría*
- ☐ **balance** — *equilibrio*

Model: *A figure is symmetric only if there is a specific line where one side is a mirror image of the other.* — A strong answer explains the line gives you something to compare the two sides across — without it, you can't test whether the sides actually match. A weak answer just says 'it looks the same.'

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **To settle the disagreement, draw a ___ through the center of each flower and check whether one side is a ___ of the other. If the test works for both, then ___ flowers have bilateral symmetry.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** symmetric
2. **Notes 2:** mirror image
3. **Notes 3:** bilateral symmetry
4. **Notes 4:** line of symmetry
5. **Notes 5:** balance
6. **Watch for this mistake:** *Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical — the mirror line can run in any direction, and the test is whether the two sides match across it.*

7. **Try It 1:** C. 46 flowers — *One side uses 14 + 9 = 23 flowers, and the mirror side uses the same 23, so the whole design uses 23 × 2 = 46 flowers.*
8. **Try It 2:**